5.2 Power of Indices

5.11 Evaluate

a) $(1+2^3)^2$

b) $\frac{2^3-1}{2^2+3}$

Solution

a) $(1+2^3)^2$

- $1+2^3=1+8=9$
- $(1+2^3)^2=9^2=81$

b) $\frac{2^3-1}{2^2+3}$

- $\frac{2^3-1}{2^2+3}$
- $2^3-1=8^2-1=7$
- $2^2+3=2^2+3=7$
- $\frac{7}{7}=1$

5.12 Simplify

$x^2y^2+3(xy)^2$

Solution

- The bracketed expression is simplified first

$(xy)^2=(xy)(xy)=x^2y^2$

- So we have

$x^2y^2+3(xy)^2=x^2y^2+3x^2y^2$

- $x^2y^2+3(xy)^2=4x^2y^2$

6.1 The laws of indices

- First law
  $a^m \times a^n = a^{m+n}$
- Second law
  $\frac{a^m}{a^n} = a^{m-n}$
- Key point: any number raised to the power 0 equals 1, that is $a^0 = 1$
- Third law
  $(a^m)^n = a^{mn}$
- Key point: $(a^m b^n)^k = a^{mk} b^{nk}$

6.2 Negative powers

- key poits - $a^{-m} = \frac{1}{a^m}$, $a^m = \frac{1}{a^{-m}}$

6.3 Square roots, cube roots and fractional powers

- Key point: $x^{\frac{1}{2}} = \sqrt{x}$, $x^{\frac{1}{3}} = \sqrt[3]{x}$
- Key point: $x^{\frac{1}{n}} = \sqrt[n]{x}$
- Key point: $a^{\frac{m}{n}} = \sqrt[n]{a^m} = (\sqrt[n]{a})^m$

6.4 Multiplication and division by powers of 10

6.5 Scientific notation

7. Simplifying algebraic expressions

7.1 Addition and substraction of like terms

7.2 Multiplying algebraic expressions and removing brackets

- Key point
  When multiplying
  positive x positive = positive
  positive x negative = negative
  negative x positive = negative
  negative x negative = positive

7.3 Removing brackets from a(b+c), a(b-c), (a+b)(c+d) and (a+b)(c-d)

- Key point
  $a(b + c) = ab + ac$
  $a(b - c) = ab - ac$

- Key point
  $(a+b)(c+d) = ac + bc + ad + bd$

8. Factorisation

8.1 Factors and common factors

8.2 Factorising quadratic expressions

- Key points

An expression of the form $ax^2 + bx + c$, where a, b and c are numbers, is called a quadratic expression. The coefficient of x^2 is a, the coefficient of x is b, and the constant term is c

- Quadratic expression where the coefficient of x^2 is 1

8.3 Different of two squares

$(x+m)(x+n) = (x+m)x + (x+m)n$
$x^2 + mx + nx + mn$
$x^2 + (m+n)x + mn$

- Expressions of the form $a^2 - b^2$ are known as the difference of two squares

9. Algebraic fractions

9.1 Introduction

When dividing

- $\frac{positive}{positive} = positive$
- $\frac{negative}{positive} = negative$
- $\frac{positive}{negative} = negative$
- $\frac{negative}{negative} = positive$

9.2 Cancelling common factors

When simplifying an algebraic fraction only factors common to both the numerator and denominator can be cancelled.

9.3 Multiplication and division of algebraic fractions

$\frac{a}{b} \times \frac{c}{d} = \frac{a \times c}{b \times d}$

9.4 Addition and subtraction of algebraic fractions

9.5. Partial fraction

Let d = highest power of the denominator
Let n = highest power of the numerator
Then
if n ≥ d the fraction is improper
if n < d the fraction is proper

10. Transposing formulae

10.1 Rearranging a formula

To transpose a formula you may:

- add the same quantity to both sides
- subtract the same quantity from both sides
- multiply or divide both sides by the same quantity
- Perform operations on both sides, such as 'square both sides', 'square root both sides' etc.

12. Sequences and series

12.1 Sequences

12.2 Arithmetic progressions

Key point

An arithmetic progression can be written

$a, a + d, a + 2d, a + 3d.....$

a is the first term, d is the commoin difference

Key point

The nth term of an arithmetic progression is givenby a + (n-1)d

12.3 Geometric progressions

Key point

A geometric progression can be written

$a, ar, ar^2, ar^3 \ldots$

a is the first term, r is the common ration

Key point

The nth term of a geometric progression is given by ar^{n-1}

12.4 Infinite sequences

12.5 Series and sigma notation

12.6 Arithmetic series

Key point

The sum of the first n terms of an arithmetic series with first term a and common difference d is denoted by S_n and given by

$S_n = \frac{n}{2}(2a + (n-1)d)$

12.7 Geometric series

Key point

The sum of the first n terms of a geometric series with first term a and common ratio r is denoted by S_n and given by

$S_n = \frac{a(1-r^n)}{1-r}$ provided r is not equal to 1

12.8 Infinite geometric series

Key point

The sum of an infinite number of terms of a geometric series is denoted by S_∞
and is given by

$S_\infty = \frac{a}{1-r}$ provided $-1 < r < 1$

13. Sets

13.1 Terminology

Key point

A set is a collection of clearly defined objects, things or states.

Key point

The order of the elements of a set is not important. There is no need to repeat elements

Key point

If all elements of a set X are also elements of a set Y then X is a subset of Y, X ⊆ Y.

Key point

The complement of X, denoted $\overline{X}$, contains all elements that are not in X.

13.2 Sets defined mathematically

Key point

The intersection of X and Y contains the elements that are common to both sets.

Key point

If $X \cap Y = \emptyset$, that is the intersection has no elements, then X and Y are disjoint sets.

Key point

The union of X and Y contains all the elements of both X and Y

13.3 Venn diagrams

13.4 Number sets

Symbol	Name	Description
$\in$	Element of	An object belongs to a set 3 ∈ {1,2,3}
$\notin$	Not an element of	An object does not belongs to a set 4 ∈ {1,2,3}
$\subseteq$	Subset	Every element of A is also in B (A can equal B)
$\subset$	Proper subset	Every element of A is in B, and B has extra elements ($A \neq B$)
$\not\subseteq$	Not a subset	A contains at least one element not in B

Symbol	Name	Description
$\supseteq$	Superset	Set A contains all elements of set B (A can equal B).
$\supset$	Proper superset	Set A contains all elements of B plus at least one extra element.
$\cup$	Union	Combines all unique elements from both A and B ("OR").
$\cap$	Intersection	Includes only elements that appear in both A and B ("AND").
$\mathcal{P}(A)$	Power Set	The set of all possible subsets of A, including $\varnothing$ and A itself.